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Explainable Multi-Label Chest X-Ray Disease Classification Using DualPool-DenseNet121 and Grad-CAM Visualization

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Abstract

The global prevalence of lung diseases has risen markedly over the past three decades, driven by factors such as population growth, harmful environmental exposures, and increased life expectancy. Epidemiological studies have highlighted not only the growing burden of chronic respiratory disorders but also the trend toward earlier onset across age groups. Early detection and timely treatment of these conditions are essential to improving long-term clinical outcomes and reducing premature mortality. Chest radiography (X-ray) remains the most widely used and cost-effective initial diagnostic tool for respiratory diseases. However, the reliable identification of multiple thoracic abnormalities, such as cardiomegaly, emphysema, hernia, infiltration, mass, nodules, atelectasis, pneumothorax, pleural thickening, pneumonia, fibrosis, consolidation, effusion and edema, poses challenges even for experienced radiologists. Recent advances in deep learning, particularly convolutional neural networks, have improved medical image analysis and classification. In this study, we propose DualPool-DenseNet121, a modified DenseNet121 architecture combining global average and global max pooling, for multi-label classification of thoracic abnormalities on the NIH ChestX-ray14 dataset. Following removal of "No Finding" cases, our analysis used 51,759 pathological radiographs from 14,402 patients, annotated for fourteen pathological conditions, partitioned patient-wise into training, validation, and test sets with verified zero patient overlap. The pipeline incorporates standardized preprocessing, offline augmentation of under-represented classes, staged transfer learning, and per-class decision thresholds selected on the validation set. Model performance was evaluated using per-class AUC together with threshold-based clinical metrics. The proposed model achieved a macro-averaged AUC of 0.803 and a micro-averaged AUC of 0.831 on the held-out test set. Zero-shot external validation on the CheXpert validation set reached a macro-averaged AUC of 0.806 across the five overlapping pathologies with sufficient positive cases. Grad-CAM visualizations indicated that, in correctly classified cases, the model frequently attended to anatomically relevant regions. These results suggest that the proposed approach provides a reliable and reproducible retrospective benchmark for multi-label thoracic abnormality classification. External validation and expert radiological assessment are required before any clinical application can be considered.

Keywords: chest X-ray; multi-label classification; explainable artificial intelligence; DenseNet121; Grad-CAM; ChestX-ray14; transfer learning; external validation

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1. Introduction

The interest in lung disease analysis using artificial intelligence methods is widely spread all over the world [1–3]. Radiologists' interpretation of radiography images remains a significant concern due to human ability to detect subtle visual features in the images, even though radiography can be completed quickly and widely due to the commonality of chest radiology imaging systems in hospitals [4]. Chest imaging (X-ray) plays a vital role in disease determination and classification as a primary clinical tool to determine lung condition. For instance, the WHO proposed in 2021 that at-risk groups be screened for tuberculosis (TB) using chest X-rays. TB is one of the leading infectious killers in the world, with over ten million cases worldwide each year [5].

Deep learning (DL) approaches offer substantial potential to improve diagnostic workflows by enabling scalable and efficient radiological assessment [6]. However, their effective use requires careful dataset preparation and interpretation due to the broad spectrum and overlapping characteristics of lung diseases [7,8]. According to the newest findings, when it comes to diagnosing problems, different machine learning (ML) techniques produce notable outcomes in classification [9,10]. Building on this, our study evaluates optimal data preparation practices for distinguishing lung diseases using X-ray datasets and DL algorithms [11,12].

The primary constraints, according to the literature, have been the large training set, the high dimensionality of the data, the independence of features, and the absence of many standard and renewed data sets [13–15]. Considerable progress has been made in the field of medical imaging, especially with the incorporation of DL technologies. It is urgent and essential to comprehend how DL is applied in this context given the increasing significance of precise and effective diagnostic procedures. X-ray imaging is known for playing a vital role in the diagnosis of disorders affecting the thoracic cavity and lungs. Regarding this, DL technologies have become key factors in interpreting diagnostic results.

There has been recent DL progress in medical imaging, but it still faces persistent barriers to wider adoption, including non-standardized preprocessing, inconsistent evaluation, and poor reproducibility [16,17]. The influence that feature engineering the conversion or fusion of various data points into new representations can have on the ultimate classification accuracy is one underestimated component of conventional ML techniques. Similarly, despite these advancements, Pillai's comprehensive review notes that the adequate interpretability and explainability of deep learning models remain a prominent challenge [17,18].

One of the essential ML techniques that is used to improve the performance and generalization capabilities of models is data augmentation (DA). Without data gathering, dataset diversity is crucial for ML training. Current data points can be changed into new data instances with the same semantic labels but slightly different looks by rotating, translating, flipping, cropping, color jittering, and other methods. This artificially broadens the training set and simulates a more generalizable learning process. Reducing overfitting, a common ML issue where models trained on rare data perform well on training data but not as well on a new dataset, so data augmentation plays a significant role before using the most suitable ML architecture. By introducing realistic transformations, models learn to ignore irrelevant variations and concentrate on patterns of decision-making [19] if augmentations can extract more information from the original dataset [20]. An in-depth analysis of DL algorithms, preprocessing techniques, datasets, evaluation metrics, a variety of applications, and their challenges, along with future research directions to address those challenges, is required for a better understanding of DL-based data augmentation for medical imaging [19]. Several types of chest radiographs can be effectively analyzed using deep learning (DL) algorithms, and their diagnostic performance can be further improved

through proper augmentation techniques that enhance dataset variability, strengthen model generalization, and reduce the risk of overfitting [21–25].

DL converts input data into outputs using models (networks) made up of a certain number of layers. A convolutional network (CN), which has several layers and uses convolution filters with a limited extent to change its input, is the most successful type of architecture model for image analysis to date [26]. Researchers from across the world have already completed many studies, with encouraging outcomes, which can support current approaches. These developments can aid in the rapid and precise identification and categorization of illnesses and offer prompt assistance in achieving remarkable outcomes in the eradication of fatal lung diseases [27]. DNN-based systems surpass traditional learning models because they become more accurate when working with massive amounts of data [28]. Through the analysis of these characteristics, the DNN can attain high accuracy on both the training dataset and standard test databases, which are identically distributed and independent of the training data (presenting the same sources of bias as the training dataset). But when it comes to real-world situations or out-of-distribution datasets, where the X-rays are collected from outside sources in comparison to the training samples, the DNNs are unable to generalize appropriately—also known as shortcut learning [29].

The COVID-19 pandemic has had a strong impact on research into the application of ML and DL in medical image analysis [30–33]. Deep neural networks (DNNs) are particularly good at classifying images, and one of the most widely used DNN classes is the convolutional neural network (CNN). The well-known CN architectures include DenseNet, Inception, ResNet, VGG, AlexNet, etc. [34]. DenseNet variants differ in block configurations, parameter counts, and depths but share a core concept: dense connectivity, where each layer receives inputs from all preceding layers. We particularly used DenseNet121 that has 121 layers and seven million parameters. A Dense Block is a fundamental component of DenseNet, made up of multiple densely connected layers. In this structure, each layer receives as input the concatenated outputs from all previous layers within the block. A key factor affecting DenseNet performance is the growth rate (k), which determines the number of new feature maps added per layer, influencing the network's ability to learn complex radiographic patterns [35]. The proposed DualPool-DenseNet121 architecture combines global average pooling and global max pooling to capture complementary feature representations. Global average pooling emphasizes globally distributed and relatively symmetric feature responses, whereas global max pooling highlights localized and potentially asymmetric pathological activations. This combination was adopted on the assumption that it would help represent both diffuse and focal radiographic manifestations; whether it does so is tested explicitly in Section 4.3 rather than assumed. Rather than proposing a new architecture, this work subjects the proposed one to explicit scrutiny: the same model and training procedure are evaluated under both partitioning strategies to measure the protocol effect directly, the contribution of the dual-pooling head is isolated in a dedicated ablation study, and the model is validated zero-shot on an independent dataset.

The remainder of this paper is organized as follows. Section 2 reviews related work on deep learning for chest radiograph classification. Section 3 describes the dataset, the preprocessing and augmentation pipeline, the proposed architecture, the transfer learning strategy and the evaluation protocol. Section 4 presents the classification results, the effect of the data partitioning strategy, the ablation study of the pooling head, the computational complexity analysis, the external validation and the Grad-CAM analysis. Section 5 discusses the findings and their limitations, and Section 6 concludes the paper.

This work presents the following key contributions:

- A reproducible pipeline for multi-label chest disease classification on ChestX-ray14, including strict patient-wise data splitting, standardized preprocessing, and offline

augmentation of under-represented classes, addressing inconsistencies found across prior studies that used heterogeneous or image-level evaluation setups.

- A modified DenseNet121 architecture, termed DualPool-DenseNet121, was built on a DenseNet121 backbone adapted to chest radiography through a staged transfer learning procedure. The architecture combines Global Average Pooling and Global Max Pooling, followed by Batch Normalization, a fully connected layer with 512 units and ReLU activation, Dropout (0.3), and a 14-unit output layer, trained with a binary cross-entropy loss. The contribution of the dual-pooling head relative to single-pooling baselines is quantified in a dedicated ablation study.
- Grad-CAM visualizations used to provide qualitative, preliminary interpretability evidence, identifying image regions associated with specific pathology predictions.
- A comprehensive evaluation using per-class AUC together with threshold-based clinical metrics (sensitivity, specificity, precision, recall, and F1), with decision thresholds selected on the validation set, and a comparison with previously published results.
- Zero-shot external validation on an independent dataset (CheXpert), performed without fine-tuning or threshold recalibration, assessing whether the model transfers across institutions, acquisition protocols, and label-generation procedures.
- An analysis of model limitations on difficult classes such as Pneumonia, Infiltration, and Consolidation, discussing dataset issues, including label noise, class imbalance, and overlapping radiographic patterns.

2. Literature Review

In recent years, several techniques, particularly advanced deep learning approaches, have been introduced to outline effective strategies for lung disease diagnosis using chest X-ray images. DenseNet-121 is usually chosen for its strong feature propagation capabilities and its effectiveness working with multilabel medical datasets [36]. According to Rajpurkar et al. (2017), a DenseNet-121 trained on the NIH ChestX-ray14 dataset achieved radiologist-level performance for pneumonia in lung disease classification [37]. Automatically detecting diseases from chest X-rays at a radiologist-level can improve healthcare, especially for communities with limited access to imaging specialists. Their proposed CheXNet (adapted DenseNet121) achieved AUC in all 14 pathologies, surpassing both Wang et al. (2017) [34] and Yao et al. (2017) [38]. For example, it obtained an AUC of 0.9248 for Cardiomegaly (compared with 0.807 and 0.904), 0.9371 for Emphysema (compared with 0.815 and 0.829), and 0.9164 for Hernia (compared with 0.767 and 0.914). Similar improvements were observed across the remaining pathologies as well [37]. Hanif et al. (2025) [36] applied DenseNet-121 on the NIH ChestX-ray14 dataset and achieved an average AUC of around 81%. Their study demonstrated that DenseNet-121, when fine-tuned and data augmentation strategies, can deliver performance in multi-label thoracic disease classification. These results further validate DenseNet-121 as a reliable backbone architecture for medical image analysis tasks, particularly in managing the complexities of class imbalance and overlapping pathologies in chest X-rays [36]. Kufel et al. [33] suggested that using DL techniques will give promising results in multi-label lung disease classification. It is possible for contemporary deep learning models to evaluate chest radiographs as effectively as radiologists. Additionally, especially in more complicated cases, they can increase radiologists' diagnostic efficiency, which directly raises the proportion of cases that are accurately diagnosed. Mann et al. used DenseNet-121 as a feature extractor, with only the fully connected layers modified during training while the rest of the architecture remained unchanged. The model achieved an AUC of 74.3% for pneumonia classification [39]. According to Chehade et al. (2025), DenseNet-121, enhanced with CBAM (convolutional block attention module) and clinical variables, proved highly effective in lung disease

classification, with the proposed preprocessing approach improving its AUC by over 5% for pneumonia and consolidation compared with baseline models [40].

Chhabra and Kumar applied a DenseNet-121 classifier to the ChestX-Ray8 dataset, comprising 108,948 frontal radiographs from 32,717 patients, and compared the resulting classification performance against radiologist assessment, reporting that the architecture was able to detect multiple thoracic conditions within a single framework [41]. Chutia et al. (2024) employed the DenseNet-121 architecture for pneumonia detection and reported improved performance, demonstrating its effectiveness in extracting deep features from chest X-ray images and enhancing classification accuracy [42]. According to Priya et al. (2022), DenseNet-121 applied to the chest X-ray dataset for multi-label disease classification reported improved results, showing the effectiveness of this architecture for automated chest pathology detection [43].

According to Shetty et al. (2025) [44] among the models tested, DenseNet-121 and its hybrid variants applied for pulmonary edema detection using the NIH ChestX-ray14 dataset DenseNet-121, particularly combined with NasNet (convolutional neural network), achieved outstanding results at a threshold of 0.5, with 94.55% accuracy, 99.27% precision, 89.77% recall, an F1-score of 0.9428, and an AUC of 0.9948. These results highlight the strong capability of DenseNet-121-based approaches in providing accurate and reliable detection of pulmonary edema from chest X-rays [44]. An et al. combined a DenseNet121 hybrid model and enhanced it with multi-head attention and dynamic pooling, which achieved 95.19% accuracy for pneumonia classification [45]. As Deep Learning (DL) approaches in artificial intelligence (AI) now play a vital role in chest X-ray disease detection, according to Muljo et al. (2022) [46], the COVID-19 Radiographic Database and the NIH chest X-ray dataset were used to explore pre-training and fine-tuning strategies on chest radiographic images. DenseNet-121 architecture was examined in detail, with a dual-dataset approach providing improved detection performance across different lung diseases and DenseNet-121 achieved an average AUC of 82.16% [46].

3. Materials and Methods

The proposed framework for multi-label chest X-ray classification consists of five main stages. The dataset preparation, image preprocessing and augmentation, feature extraction using the proposed DualPool-DenseNet121 architecture, model training and explainability analysis using Grad-CAM. The following subsections describe each stage in detail. All experiments were conducted on a dedicated GPU server at Vilnius Gediminas Technical University (VILNIUS TECH, Vilnius, Lithuania), running Ubuntu 24.04.4 LTS and equipped with an NVIDIA B200 GPU (183 GB, NVIDIA Corporation, Santa Clara, CA, USA) with driver version 580.126.16, CUDA 13.0 and cuDNN 9.19.0, and an Intel Xeon Platinum 8570 CPU (Intel Corporation, Santa Clara, CA, USA). Models were implemented in Python 3.13.13 (Python Software Foundation, Wilmington, DE, USA) using PyTorch 2.11.0 and torchvision 0.26.0 (The Linux Foundation, San Francisco, CA, USA), executed within a JupyterLab environment. Grad-CAM visualizations were produced using the open-source grad-cam library version 1.5.5 (J. Gildenblat, https://github.com/jacobgil/pytorch-grad-cam, accessed on 7 February 2026). Data processing, evaluation and visualization were performed using NumPy 2.4.3, SciPy 1.17.1, pandas 3.0.2, scikit-learn 1.8.0, OpenCV 4.13.0 and Matplotlib 3.10.8. Inference time was measured on this configuration (1.22 ms per image on GPU, 364.8 ms on CPU).

3.1. The Dataset

We used the NIH ChestX-ray14 dataset, openly accessible on Kaggle and released by the National Institutes of Health (NIH) [47]. Classification of this dataset is a multi-label

problem: it is highly imbalanced, and a single patient may present multiple co-occurring diseases. Each radiograph is annotated for up to fourteen pathologies (Figure 1): atelectasis, cardiomegaly, consolidation, edema, effusion, emphysema, fibrosis, hernia, infiltration, mass, nodule, pleural thickening, pneumonia, and pneumothorax. The reported accuracy of the dataset labels is greater than 90% [34]. The resulting patient-wise partitioning is summarized in Table 1. From the 51,759 pathological radiographs (14,402 patients), the data were split into training (36,144 original images, 10,081 patients), validation (7775 images, 2160 patients), and test (7840 images, 2161 patients) subsets. Offline augmentation of minority classes increased the training set to 118,227 images, while the validation and test sets retained original images only.

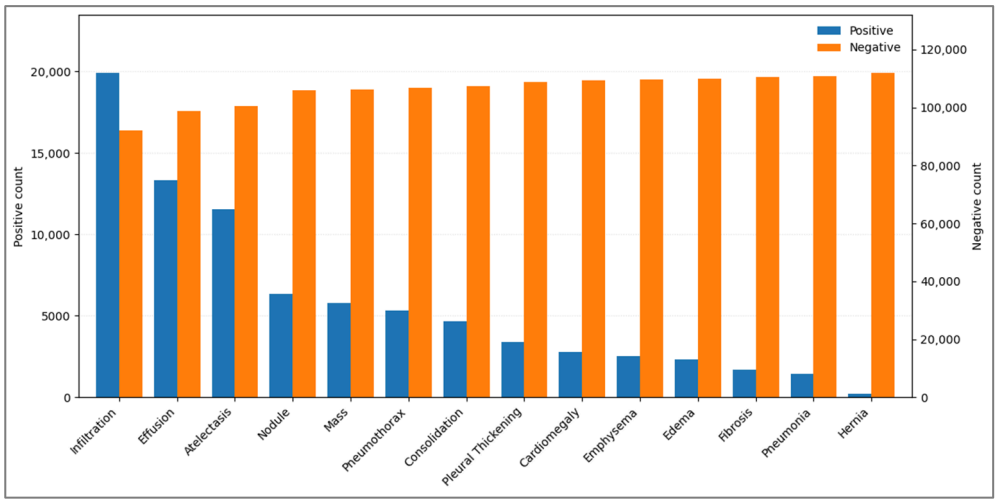


Figure 1. Dataset distribution across all fourteen different diseases.

Table 1. Patient-wise partitioning of the dataset after exclusion of "No Finding" cases. Offline augmentation was applied only to minority classes within the training set; validation and test sets contain original images only.

Subset	Patients	Original Images	Augmented Images	Total Images
Training	10,081	36,144	82,083	118,227
Validation	2160	7775	-	7775
Test	2161	7840	-	7840
Total	14,402	51,759	82,083	133,842

A patient-wise splitting strategy was adopted so that no patient appeared in more than one subset, eliminating patient- and image-level leakage across partitions; zero patient overlaps were verified. Stratification was applied to maintain proportional representation of positive and negative cases across subsets, ensuring consistent class distributions during training and evaluation. The test set remained completely unseen during model development and was used only for the final performance assessment.

3.2. Data Preprocessing and Augmentation

To manage transformations that generate empty regions, the nearest-neighbor fill mode was used. Images were resized to 320 × 320 pixels and processed in 32 batches, with a fixed random seed (SEED = 1337) to ensure reproducibility. Before training, a consistency check was performed to verify that every metadata entry had a valid corresponding image file and reliable labels. Entries with abnormal or unreliable labels (bad_labels) were removed, reducing the original 112,120 radiographs of ChestX-ray14 to 111,863. Radiographs annotated exclusively as "No Finding" were then excluded, since this study

addresses multi-label classification of pathological findings, yielding the 51,759 pathological radiographs from 14,402 patients that were partitioned patient-wise as summarized in Table 1.

Augmentation was applied to increase the model's robustness and generalization. Each radiograph was first sample-wise centered and normalized, and mild geometric and photometric transformations were then applied; the complete set of parameters is given in Table 2. Horizontal flipping was disabled because left–right inversion alters the anatomical interpretation of chest radiographs (Figure 2). This pipeline increased dataset diversity while preserving clinical relevance, reducing overfitting and enabling the model to focus on disease-related patterns rather than irrelevant variations.

Table 2. Augmentation parameters applied to the training set.

Transformation	Setting
Sample-wise centering and normalization	Zero mean
Rotation	±10°
Width and height shift	±10%
Zoom	0.9–1.1
Brightness	0.8–1.0
Horizontal flip	Disabled
Fill mode	Nearest-neighbor

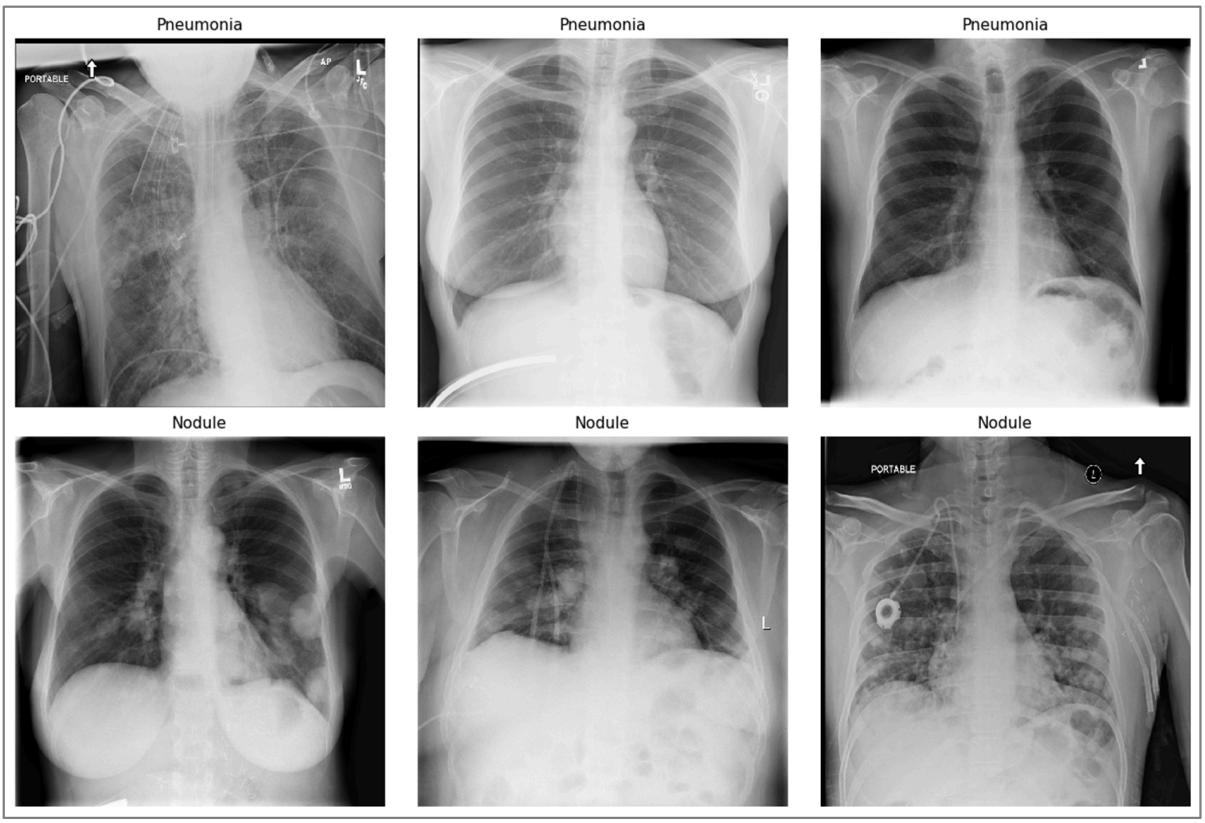


Figure 2. Example of augmented chest X-ray images from the ChestX-ray Dataset representing different thoracic diseases.

3.3. Proposed DualPool-DenseNet121 Architecture

With the growing depth of convolutional neural networks, a key challenge emerged: information and gradients tended to vanish as they propagated through multiple layers. DenseNet addresses this by introducing dense connections, where each layer receives

the feature maps of all preceding layers and contributes its own to all subsequent ones. This design ensures effective gradient flow, strengthens feature propagation, encourages feature reuse, and reduces the number of parameters compared with traditional deep architectures [48]. The proposed architecture employs DenseNet-121 as the backbone feature extractor, with the original classification layer removed to enable task-specific adaptation. The backbone was initialized from ImageNet-pretrained weights and subsequently adapted through a domain-specific checkpoint (Section 3.4). The proposed framework for multi-label chest X-ray classification consists of five main stages: dataset preparation, image preprocessing and augmentation, feature extraction using the proposed DualPool-DenseNet121 architecture, model training, and explainability analysis using Grad-CAM. The complete experimental workflow is summarized in Figure 3. The following subsections describe each stage in detail.

The combination of Global Average Pooling (GAP) and Global Max Pooling (GMP) allows the architecture to learn both globally distributed and localized feature representations. This design is particularly relevant for chest X-ray analysis, where symmetric anatomical structures coexist with asymmetric disease manifestations, requiring the model to capture both global and local radiographic characteristics. Given the DenseNet-121 feature maps X, the two pooling operations are applied in parallel and concatenated:

$$F = Concatenate(GAP(X), GMP(X)) \quad (1)$$

where X denotes the DenseNet-121 feature maps, and GAP(X) and GMP(X) each produce a 1024-dimensional vector, yielding a 2048-dimensional pooled representation F.

This concatenated vector is passed to a custom classification head consisting of Batch Normalization, a fully connected layer with 512 units and ReLU activation, Dropout (0.3), and a final fully connected layer with 14 output units, one per thoracic pathology. Raw logits are produced at the output; class probabilities are obtained via a sigmoid function applied within the binary cross-entropy loss during training (Section 3.5), enabling independent probability estimation for each class in the multi-label setting. The final model thus combines the DenseNet-121 convolutional backbone with this dual-pooling classification head (Figure 4).

To further investigate interpretability, Grad-CAM visualizations were incorporated to identify image regions contributing to specific disease predictions.

Figure 4 illustrates the proposed DualPool-DenseNet121 model architecture, which combines an ImageNet-pretrained DenseNet121 backbone with a custom dual-pooling classification head. The backbone consists of four dense blocks with 6, 12, 24, and 16 layers, respectively. The network begins with an initial 7×7 convolutional layer, followed by a 3×3 max pooling operation to reduce spatial dimensions. This design enables efficient gradient propagation, improved feature reuse, and reduced parameter count compared with traditional CNNs.

The output feature maps are aggregated using Global Average Pooling and Global Max Pooling, followed by Batch Normalization, a fully connected layer with 512 units and ReLU activation, and Dropout (0.3). The final fully connected layer contains 14 output nodes, each corresponding to one thoracic pathology (Atelectasis, Cardiomegaly, etc.). A sigmoid activation function is applied at the output to support multi-label classification, allowing the model to predict the presence of multiple diseases in a single chest X-ray image.

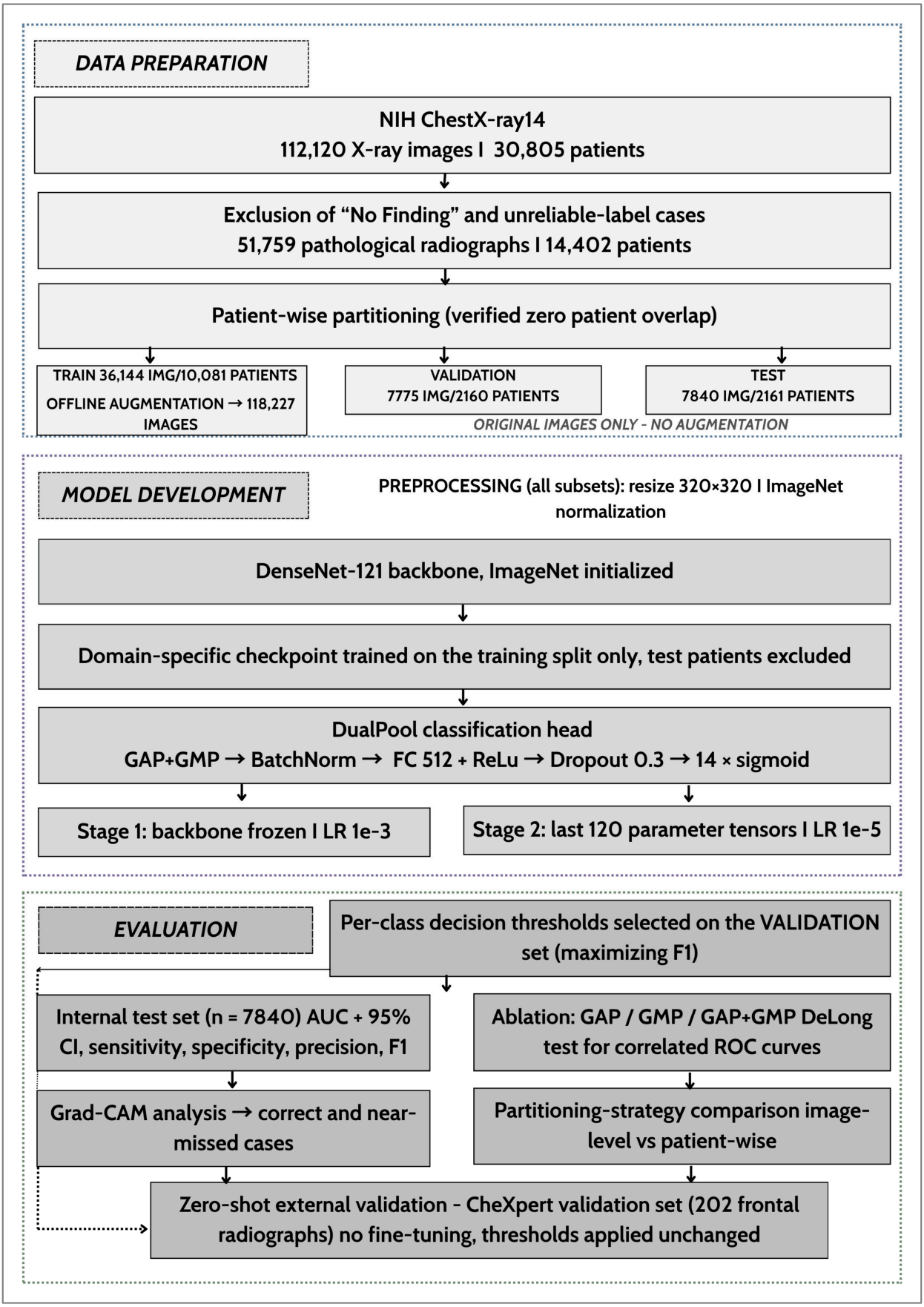


Figure 3. Overview of the experimental workflow.

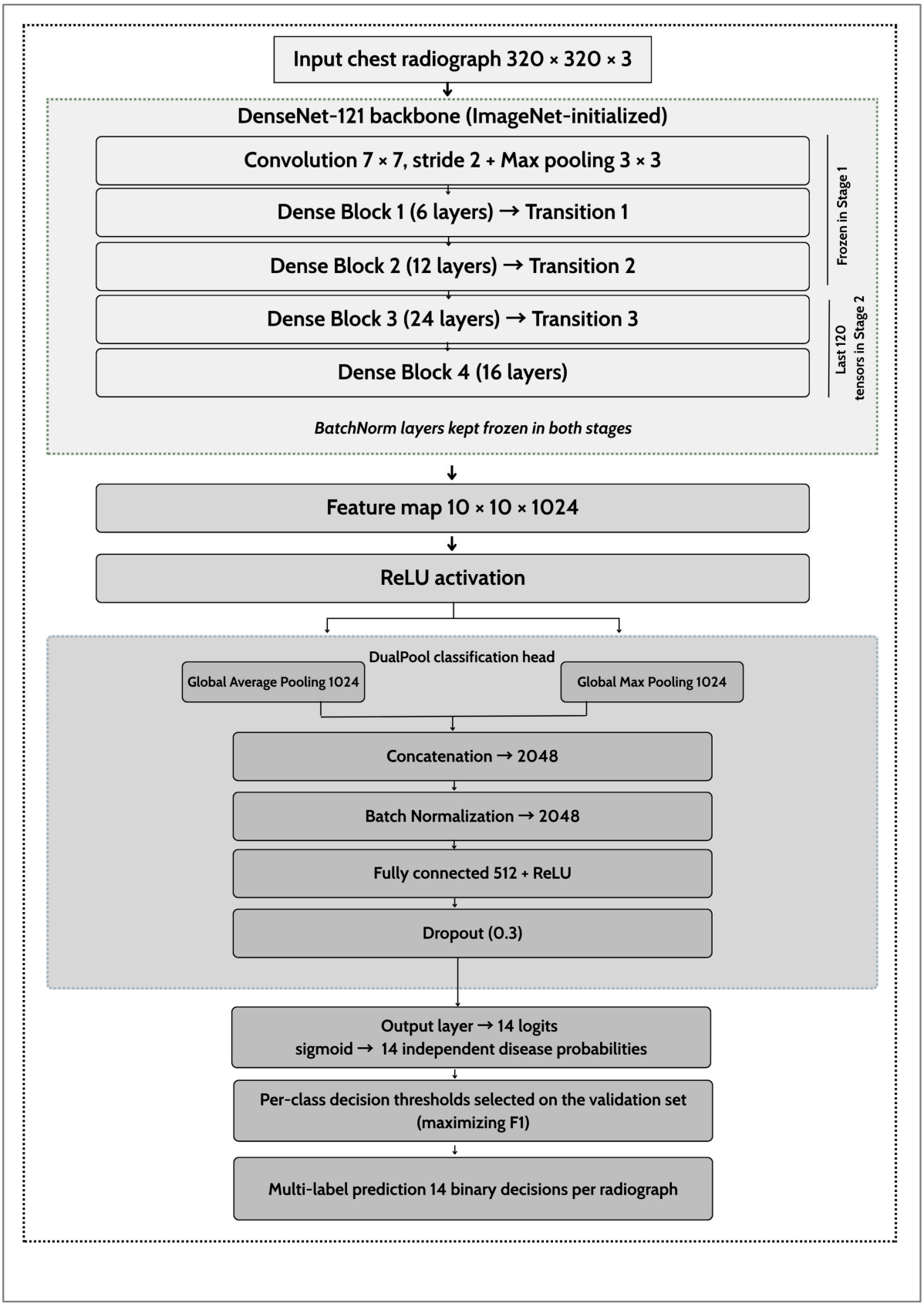


Figure 4. Overview of the proposed DualPool-DenseNet121 framework.

3.4. Transfer Learning Strategy

Transfer learning was employed to improve convergence and to leverage previously learned radiographic representations. The proposed DualPool-DenseNet121 was built upon a DenseNet-121 backbone (ImageNet-pretrained) with the original classifier replaced by the dual-pooling head described in Section 3.3. To adapt the backbone to chest radiography, the network was first initialized from a domain-specific checkpoint obtained from a preliminary training stage conducted exclusively on the training split (the first stage described in Section 3.5). Identical patient-wise train, validation and test partitions were used across all training stages; the test set was held out from the preliminary checkpoint training and from all subsequent fine-tuning. Patient-level separation was enforced so that no patient appeared in more than one subset at any stage, and zero patient overlap across the three partitions was verified.

Fine-tuning proceeded in two stages. The backbone was initially frozen while the new classification head was trained. Subsequently, the last 120 parameter tensors of the backbone were unfrozen and fine-tuned with a reduced learning rate of 1×10^{-5} using the AdamW optimizer (PyTorch 2.11.0, The Linux Foundation, San Francisco, CA, USA), while all Batch Normalization layers were kept frozen to prevent drift in the batch statistics during fine-tuning. This staged strategy improved task-specific adaptation while reducing overfitting and preserving previously learned representations.

3.5. Model Training

The model was trained in two stages. In the initial stage, the DenseNet-121 backbone was frozen and only the custom classification head was trained. Since chest disease detection was formulated as a multi-label classification problem, a sigmoid activation function was applied independently to each output neuron to estimate the probability of each pathology:

$$\sigma(x) = \frac{1}{1 + {}^{-x}} \tag{2}$$

where x denotes the logit produced by the network and $\sigma(x)$ denotes the predicted probability of a given pathology. The sigmoid maps each output to the interval [0, 1], allowing probabilities to be estimated independently for every disease category. This is appropriate for multi-label classification, where several thoracic pathologies may be present simultaneously in a single radiograph.

Optimization was performed with the AdamW optimizer using a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4}. Because the pathology prevalence in ChestX-ray14 varies by two orders of magnitude, a weighted binary cross-entropy loss was used, in which the contribution of positive examples of each class is scaled by a class-specific weight:

$$L = -\left(\frac{1}{14}\right)\sum_{i=1}^{14}[w_i y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{3}$$

where $y_i \in \{0, 1\}$ is the reference label for the i-th pathology, p_i the corresponding predicted probability, and w_i the positive-class weight derived from the frequency of that pathology in the training set. The loss is computed independently for each of the fourteen categories and averaged to obtain the training objective.

The batch size was 32 and training was allowed to run for up to 15 epochs. Performance was monitored on the validation set using the macro-averaged multi-label AUC. Three mechanisms were used to stabilize optimization and limit overfitting:

- the learning rate was reduced by a factor of 0.3 when the validation macro-AUC did not improve for two consecutive epochs;

- training was stopped when the validation macro-AUC did not improve for four consecutive epochs;
- the parameters achieving the highest validation macro-AUC were retained, and gradients were clipped to a maximum norm of 1.0 at every update.

In the second stage, the last 120 parameter tensors of the backbone were unfrozen for fine-tuning. All Batch Normalization layers were kept frozen to avoid drifting in the batch statistics during fine-tuning, while the remaining unfrozen parameters were updated. Fine-tuning used a reduced learning rate of 1×10^{-5} with the same optimizer, batch size, loss function, monitoring criterion and stopping rules as the first stage. Both stages used the same patient-wise training split; no validation or test data were used for parameter updates at any point. All hyperparameters used are summarized in Table 3.

Table 3. Training hyperparameters for the two-stage transfer learning procedure.

Parameter	Stage 1 Head Training	Stage 2 Fine-Tuning
Input resolution	320×320	320×320
Backbone	DenseNet121, frozen (feature extractor)	Last 120 parameter tensors unfrozen
Batch Normalization layers	Frozen	Frozen
Classification head	Trained	Trained
Dropout	0.3	0.3
Optimizer	AdamW	AdamW
Learning rate	1×10^{-3}	1×10^{-5}
Weight decay	1×10^{-4}	1×10^{-4}
Loss function	Weighted binary cross-entropy	Weighted binary cross-entropy
Batch size	32	32
Epochs	15	15
Learning rate scheduler	ReduceLROnPlateau (factor 0.3, patience 2)	ReduceLROnPlateau (factor 0.3, patience 2)
Early stopping	Patience 4, monitored on validation macro-AUC	Patience 4, monitored on validation macro-AUC
Model checkpointing	Best validation AUC	Best validation AUC
Monitored metric	Multi-label AUC (validation)	Multi-label AUC (validation)
Hardware	NVIDIA B200 GPU	NVIDIA B200 GPU

3.6. Class Imbalance Mitigation

The ChestX-ray14 dataset is highly imbalanced, with substantial variation in the prevalence of thoracic pathologies. Common findings such as Infiltration and Effusion are heavily over-represented, whereas conditions such as Hernia and Pneumonia are comparatively rare. Such imbalance can bias training towards majority classes and obscure performance on infrequent conditions.

We addressed this imbalance through two complementary strategies. First, patient-wise stratified partitioning preserved the relative proportions of positive and negative cases for each pathology across the training, validation, and test subsets, ensuring consistent class distributions during both training and evaluation. Second, to reduce the under-representation of rare pathologies in the training set, we applied targeted offline augmentation to minority classes, generating additional augmented radiographs so that under-represented conditions contributed more substantially to the training objective. In addition, a class-weighted binary cross-entropy loss was employed during both stages, assigning higher weights to under-represented pathologies; to prevent training instability for extremely rare classes, these weights were capped at a fixed maximum value of 10.

It should be noted that stratified partitioning alone does not alter the underlying class distribution; it preserves representativeness across subsets but does not resolve imbalance during learning. The augmentation and loss-weighting strategies described above were, therefore, introduced specifically to mitigate this effect.

3.7. Performance Evaluation

This section explains the evaluation phase. Two configurations were compared: a baseline DenseNet121 initialized with ImageNet pretrained weights and the proposed DualPool-DenseNet121 initialized with domain-specific pretrained weights. The AUC (Area Under the Curve) of the ROC (Receiver Operating Characteristic) is a standard metric for evaluating a model's ability to distinguish between classes across all thresholds. A higher AUC reflects stronger overall performance, with curves closer to the top-left corner indicating superior classification capability. This metric is widely adopted by researchers worldwide for analyzing classification outcomes and comparing model performance [49,50].

$$AUC = \int_0^1 TPR(FPR)d(FPR) \tag{4}$$

where:

$$TPR = \frac{TP}{TP + FN} \tag{5}$$

$$FPR = \frac{FP}{FP + TN} \tag{6}$$

TP (True Positive) denotes correctly classified positive cases, TN (True Negative) denotes correctly classified negative cases, FP (False Positive) represents negative cases incorrectly classified as positive, and FN (False Negative) represents positive cases incorrectly classified as negative. The True Positive Rate (TPR), also known as sensitivity or recall, measures the proportion of actual positive cases correctly identified by the model, while the False Positive Rate (FPR) measures the proportion of actual negative cases incorrectly classified as positive [50].

4. Results

This section presents the performance results of the evaluated models on the ChestX-ray14 dataset. The comparison is based on per-class and mean AUC scores, allowing assessment of the effectiveness of the proposed DualPool-DenseNet121 architecture relative to the baseline DenseNet121 model.

4.1. Classification Performance Analysis

This subsection presents the classification performance achieved by the proposed DualPool-DenseNet121 model on the NIH ChestX-ray14 dataset. Model effectiveness was evaluated using Receiver Operating Characteristic (ROC) curve analysis and the Area Under the Curve (AUC) metric, which quantifies the ability of the model to rank positive cases above negative ones independently of any decision threshold. Because the task is multi-label and the pathology prevalence varies by two orders of magnitude, AUC is aggregated in two complementary ways. The macro-averaged AUC is obtained by computing the AUC separately for each of the fourteen pathologies and taking the unweighted mean, so that every pathology contributes equally regardless of how many positive cases it contains. The micro-averaged AUC pools the predictions of all pathologies into a single ranking problem before computing the AUC and is, therefore, dominated by the most prevalent classes. The macro average is the more demanding of the two, as it gives rare pathologies such as Hernia (37 positive cases) the same weight as Infiltration

(2920 positive cases); the micro average better reflects the performance on a case mix resembling the test distribution. Both are reported throughout, and per-class values are given in Table 4.

Table 4. Per-class and macro-averaged AUC comparison between the baseline DenseNet121 and the proposed DualPool-DenseNet121 model. Confidence intervals are reported for the proposed model and were obtained by bootstrap resampling of the test set with 2000 replicates. Prevalence denotes the proportion of positive cases in the test set and defines the random baseline for PR-AUC, which is reported for the proposed model.

Disease	Prevalence	DenseNet121	DualPool-DenseNet121	95% CI	Ratio	PR-AUC
Atelectasis	0.240	0.745	0.776	[0.764, 0.788]	2.2	0.535
Cardiomegaly	0.046	0.797	0.898	[0.880, 0.915]	9.1	0.418
Consolidation	0.094	0.660	0.737	[0.719, 0.756]	2.3	0.217
Edema	0.040	0.790	0.853	[0.829, 0.875]	7.1	0.283
Effusion	0.258	0.775	0.836	[0.826, 0.846]	2.6	0.658
Emphysema	0.057	0.849	0.916	[0.901, 0.930]	10.5	0.599
Fibrosis	0.038	0.743	0.812	[0.786, 0.838]	5.0	0.187
Hernia	0.005	0.828	0.885	[0.831, 0.934]	24.4	0.115
Infiltration	0.372	0.680	0.706	[0.694, 0.717]	1.5	0.571
Mass	0.117	0.740	0.800	[0.782, 0.817]	4.1	0.476
Nodule	0.120	0.679	0.772	[0.755, 0.788]	3.2	0.385
Pleural Thickening	0.062	0.720	0.753	[0.732, 0.773]	2.9	0.180
Pneumonia	0.030	0.590	0.675	[0.642, 0.709]	1.9	0.057
Pneumothorax	0.095	0.818	0.827	[0.811, 0.842]	3.7	0.352
Macro-averaged AUC	0.112	0.743	0.803	[0.797, 0.810]	0.32	0.360

Figure 5 shows that the proposed DualPool-DenseNet121 model reaches a macro-averaged AUC of 0.803 and a micro-averaged AUC of 0.831, with the strongest per-class results for Emphysema (0.916), Cardiomegaly (0.898), Hernia (0.885), Edema (0.853) and Effusion (0.836). More difficult cases, such as Pneumonia (0.675) and Infiltration (0.706), remained challenging. Detailed per-class results and the comparison with the standard DenseNet121 baseline are provided in Table 4.

The 95% confidence intervals indicate that per-class AUC estimates are precise for prevalent pathologies, with intervals spanning less than 0.03 for Infiltration, Effusion and Atelectasis, whereas rare classes are estimated less precisely: the interval for Hernia (37 positive cases) spans 0.103. The macro-averaged AUC of 0.803 has a confidence interval of [0.797, 0.810].

Because ROC-AUC is insensitive to the number of negative cases, precision–recall AUC is additionally reported for each pathology (Table 4). Since the baseline of this metric equals the class prevalence rather than 0.5, the ratio of PR-AUC to prevalence is also given. The macro-averaged PR-AUC of 0.360 against a prevalence baseline of 0.112 corresponds to a threefold improvement overall, and the ratio is highest for the rarest pathologies, reaching 24.4 for Hernia, whereas it falls to 1.5 for Infiltration, the most prevalent class. Discriminative ability in the rare classes is, therefore, not an artefact of their low prevalence.

Compared with the baseline DenseNet121 model initialized with ImageNet weights, the proposed DualPool-DenseNet121 architecture achieves higher AUC across all fourteen pathologies, with the largest gains observed for Cardiomegaly (+0.101), Nodule (+0.093), Pneumonia (+0.085) and Consolidation (+0.077). The two models differ in two respects: the dual-pooling classification head and the staged domain-specific initialization. Their individual contributions are isolated in the ablation study reported in Section 4.3.

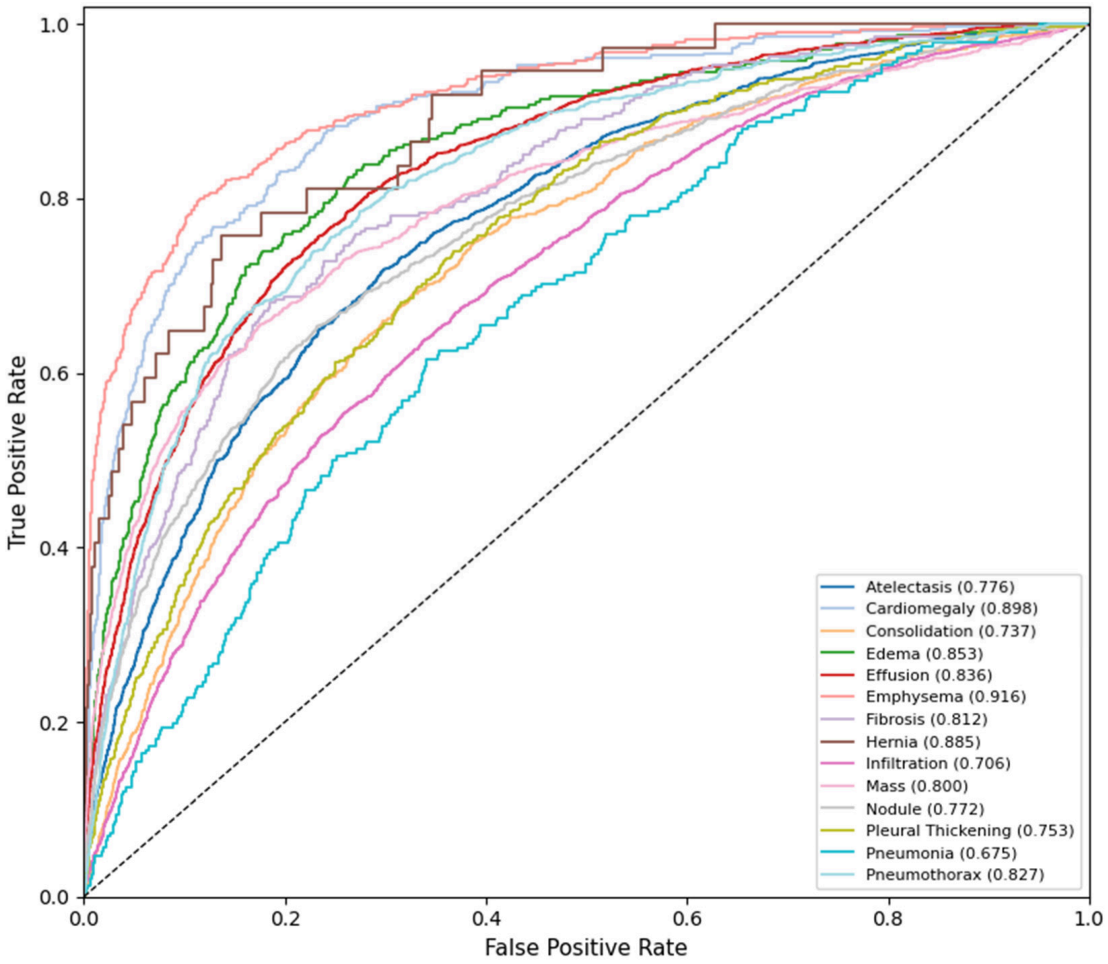


Figure 5. ROC curves for 14 lung diseases obtained using the proposed DualPool-DenseNet121 model.

Because the dataset is highly imbalanced and a fixed cutoff of 0.5 is not appropriate for multi-label classification with widely varying prevalence, a separate decision threshold was selected for each pathology. For every class, the threshold maximizing the F1-score on the validation set was retained and subsequently applied unchanged to the test set. No threshold was tuned on test data at any point. This procedure explains why some correctly classified cases have predicted probabilities below 0.5. The resulting per-class thresholds are reported alongside the corresponding performance metrics in Table 5.

$$t_c* = arg\ max\ F1\ (\ yc,\ val\ ,\ 1[\ pc,\ val\ \geq\ t\ ]\ ),\ \ t\ \in\ \{\ 0.05,\ 0.06,\ \ldots,\ 0.95\ \} \qquad (7)$$

where the threshold t_c^* is obtained for each pathology on the validation set and was applied unchanged to the test set and to the external validation set. No threshold was tuned on test data at any stage.

Table 5. Per-class classification performance of DualPool-DenseNet121 on the held-out test set (7840 radiographs from 2161 patients). Decision thresholds were selected independently for each pathology on the validation set using F1 statistics and applied unchanged to the test set. Sensitivity and recall are numerically equivalent and are both reported for completeness. TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives at the stated threshold.

Disease	AUC	Threshold	Sensitivity	Specificity	Precision	Recall	F1	TP	TN	FP	FN
Atelectasis	0.776	0.44	0.615	0.787	0.477	0.615	0.537	1156	4691	1268	725
Cardiomegaly	0.898	0.80	0.489	0.972	0.457	0.489	0.472	176	7271	209	184

Table 5. *Cont.*

Disease	AUC	Threshold	Sensitivity	Specificity	Precision	Recall	F1	TP	TN	FP	FN
Consolidation	0.737	0.62	0.528	0.804	0.218	0.528	0.308	388	5712	1393	347
Edema	0.853	0.64	0.462	0.944	0.254	0.462	0.328	144	7106	422	168
Effusion	0.836	0.60	0.720	0.801	0.557	0.720	0.628	1454	4664	1156	566
Emphysema	0.916	0.81	0.554	0.983	0.662	0.554	0.603	247	7268	126	199
Fibrosis	0.812	0.69	0.193	0.983	0.311	0.193	0.238	57	7419	126	238
Hernia	0.885	0.21	0.243	0.995	0.180	0.243	0.207	9	7762	41	28
Infiltration	0.706	0.24	0.826	0.433	0.464	0.826	0.594	2413	2132	2788	507
Mass	0.800	0.65	0.488	0.932	0.490	0.488	0.489	449	6451	468	472
Nodule	0.772	0.55	0.499	0.873	0.349	0.499	0.411	470	6021	878	471
Pleural Thickening	0.753	0.55	0.332	0.912	0.200	0.332	0.250	162	6704	648	326
Pneumonia	0.675	0.33	0.164	0.941	0.078	0.164	0.106	38	7160	448	194
Pneumothorax	0.827	0.84	0.414	0.939	0.417	0.414	0.415	307	6668	430	435
Macro-averaged AUC	0.803	0.569	0.466	0.878	0.365	0.466	0.399	-	-	-	-
Micro-averaged AUC	0.831	-	-	-	-	-	-	-	-	-	-

Table 6 compares the proposed model with previously published results on ChestX-ray14 under matching partitioning strategies.

Table 6. Comparison with previously published results on ChestX-ray14. Only studies using patient-wise data partitioning are included, since random image-level splits allow radiographs of the same patient to appear in both training and test sets and yield optimistically biased estimates. For Hanif et al., both the binary cross-entropy baseline and the proposed focal ranking loss are reported; values are those obtained without test-time augmentation.

Disease	Patient-Wise Partitioning				Image-Level Partitioning	
Author	Mao et al. (2018) [51]	Hanif et al. (2025) BCE [36]	Hanif et al. (2025) FZLPR [36]	This Work	Kufel et al. (2023) [33]	This Work
Atelectasis	0.750	0.772	0.773	0.776	0.817	0.821
Cardiomegaly	0.869	0.874	0.879	0.898	0.911	0.890
Consolidation	0.728	0.746	0.745	0.737	0.879	0.828
Edema	0.834	0.841	0.840	0.853	0.898	0.937
Effusion	0.810	0.827	0.827	0.836	0.879	0.901
Emphysema	0.870	0.907	0.908	0.916	0.935	0.928
Fibrosis	0.798	0.800	0.806	0.812	0.824	0.857
Hernia	0.878	0.728	0.891	0.885	0.890	0.968
Infiltration	0.687	0.683	0.666	0.706	0.716	0.695
Mass	0.782	0.824	0.825	0.800	0.853	0.833
Nodule	0.726	0.770	0.758	0.772	0.771	0.763
Pleural Thickening	0.758	0.771	0.767	0.753	0.812	0.841
Pneumonia	0.695	0.704	0.716	0.675	0.769	0.835
Pneumothorax	0.845	0.863	0.864	0.827	0.898	0.912
Macro-averaged AUC	0.788	0.794	0.805	0.803	0.847	0.858

Under patient-wise partitioning, the proposed model achieves a macro-averaged AUC of 0.803, with a 95% confidence interval of [0.797, 0.810] obtained by bootstrap resampling of the test set. The published results in this group are 0.788 (Mao et al. [51]) and 0.794 and 0.805 (Hanif et al. [36], for binary cross-entropy and focal ranking loss, respectively). The two values reported by Hanif et al. fall inside the interval of the present model, as confidence intervals are not reported in either published study; a formal test of the differences is not possible.

Across this group, the proposed model gives the highest per-class AUC for eight pathologies: Atelectasis, Cardiomegaly, Edema, Effusion, Emphysema, Fibrosis, Infiltration and Nodule. It is lower for Consolidation, Mass, Pleural Thickening, Pneumonia and

Pneumothorax, and for Hernia it exceeds the binary cross-entropy result but not the focal ranking result.

Under image-level partitioning, the model achieves 0.858 compared with 0.847 reported by Kufel et al. [33], with higher per-class AUC for eight pathologies: Atelectasis, Edema, Effusion, Fibrosis, Hernia, Pleural thickening, Pneumonia and Pneumothorax.

The difference of 0.055 between the two configurations of the same model exceeds the difference within any of the published results compared here, which illustrates that the partitioning strategy has a larger influence on reported performance than the differences between the methods being compared. Comparisons across studies are, therefore, informative only when the partitioning strategy is matched, and results reported without a stated strategy cannot be placed on either side of this table with confidence. This grouping is consistent with the observation of Hanif et al. [36], who note that studies not using the official split of ChestX-ray14 report higher average AUC and that such comparisons must, therefore, be interpreted with caution.

4.2. Effect of the Partitioning Strategy

To quantify the influence of data partitioning, the same architecture and training protocol were evaluated under two splitting strategies. Under a random image-level split, in which radiographs of the same patient may appear in more than one subset, the model achieved a macro-averaged AUC of 0.858 (Figure 6). Under strict patient-wise partitioning, verified to contain no patient overlap, the same model achieved 0.803, an absolute difference of 0.055.

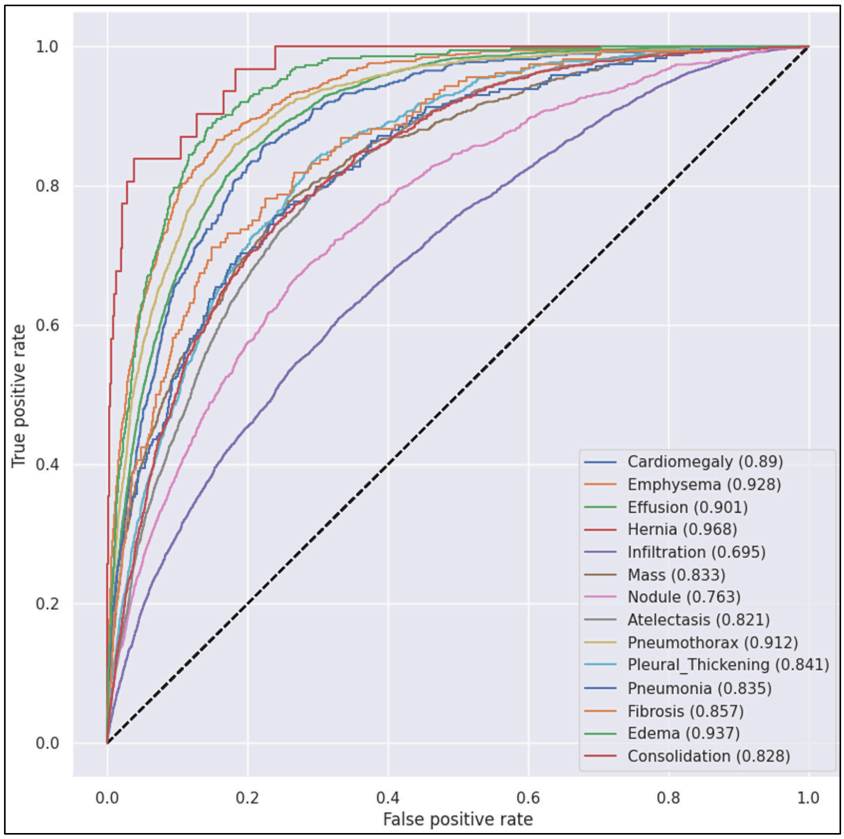


Figure 6. ROC curves for the fourteen pathologies obtained under a random image-level split, in which radiographs of the same patient may appear in more than one subset. The per-class AUC values shown here (macro average 0.858) are optimistically biased and are presented only to quantify the effect of the partitioning strategy.

This difference is consistent with previously reported effects of partitioning strategy on this dataset, where absolute gains of 3.4 percentage points have been observed for a random patient-wise split and 10.8 percentage points for an image-wise split [52]. ChestX-ray14 contains an average of 3.6 radiographs per patient, so image-level partitioning almost inevitably places correlated images of the same patient in both training and test sets. All results reported in this work use the patient-wise split, and comparisons in Table 6 are restricted to studies employing the same protocol.

4.3. Ablation Study of the Pooling Strategy

To isolate the contribution of the dual-pooling head, three variants of the classification head were trained under a separate but mutually identical protocol, differing only in the pooling operation. As shown in Table 7, the three variants achieved macro-averaged AUC values within 0.004 of each other. Per-class differences were assessed using DeLong's test for correlated ROC curves, which accounts for the fact that the variants are evaluated on the same test set and their predictions are, therefore, not independent; differences were statistically significant for 4 of 14 pathologies against GAP and 3 of 14 against GMP, with no consistent direction.

Table 7. Ablation study of the pooling strategy. Statistical significance of per-class differences was assessed using DeLong's test for correlated ROC curves.

Pooling Strategy	Macro-Averaged AUC	Significant Differences vs. GAP + GMP
GAP only	0.795	4/14
GMP only	0.791	3/14
GAP + GMP (proposed)	0.793	-

These results indicate that combining average and maximum pooling does not confer a statistically robust advantage over global average pooling alone on this dataset. We, therefore, report the dual-pooling head as a design choice with performance comparable to standard pooling, rather than as the source of the improvement over the baseline reported in Table 4. That improvement is instead attributable to the staged domain-specific initialization, the targeted augmentation of under-represented classes, and the per-class threshold selection described in Sections 3.2, 3.4 and 3.7.

4.4. Computational Complexity

The proposed DualPool-DenseNet121 comprises 8.01 M parameters, of which approximately 7.0 M belong to the DenseNet-121 backbone and 1.05 M to the classification head, corresponding to a stored model size of 32.1 MB, and requires 5.91 GMACs (11.82 GFLOPs) per 320 $\times$ 320 input image. The head is larger than a standard single-pooling head because concatenating global average and global max pooling produces a 2048-dimensional representation rather than 1024, doubling the input dimension of the first fully connected layer; the pooling operations themselves introduce no learnable parameters. The arithmetic cost is dominated by the convolutional backbone, so the dual-pooling design increases the parameter count without a corresponding increase in computation.

The average inference time was 1.22 ms per image on an NVIDIA B200 GPU, measured with a batch size of 32 after discarding warm-up iterations, corresponding to a throughput of 819 images per second. On the same server, inference on CPU (Intel Xeon Platinum 8570) required 364.8 ms per image. These figures describe the forward pass alone. End-to-end processing of the test set, including reading and decoding the original radiographs from disk, resizing and normalization, took 80 ms per image, so the complete pathological subset of 51,759 radiographs would be processed in approximately one hour on a single GPU. The

cost of a deployed system is, therefore, dominated by data handling rather than by the model itself.

4.5. External Validation

All results reported were obtained on a held-out subset of the same dataset used for training, and therefore, share its acquisition protocol, patient population and label-generation procedure. The set contained 202 frontal radiographs from 200 patients, labelled by consensus of three board-certified radiologists. No fine-tuning or threshold recalibration was performed. Internal AUC refers to the NIH ChestX-ray14 test set (7840 radiographs). Confidence intervals were obtained by bootstrap resampling with 2000 replicates. Macro averages are computed over the five classes with at least 30 positive cases. To assess transferability, we performed zero-shot external validation on the CheXpert validation set [53], which comprises 234 radiographs from 200 patients annotated by consensus of three board-certified radiologists. Restricting the evaluation to frontal projections, consistent with the training data, 202 radiographs and seven pathologies overlap between the two label sets. No fine-tuning was performed and no thresholds were recalibrated. Per-class results with confidence intervals are presented in Table 8.

Table 8. Zero-shot external validation on the CheXpert validation set.

CheXpert Label	NIH Label	Positive Cases	Internal AUC	External AUC	95% CI (External)	Difference
Atelectasis	Atelectasis	75	0.776	0.753	[0.684, 0.817]	−0.023
Cardiomegaly (Enlarged Cardiomediastinum *)	Cardiomegaly	66	0.898	0.785	[0.714, 0.847]	−0.114
Pleural Effusion	Effusion	64	0.836	0.880	[0.823, 0.926]	+0.044
Edema	Edema	42	0.853	0.732	[0.644, 0.812]	−0.121
Consolidation	Consolidation	32	0.737	0.878	[0.810, 0.937]	+0.140
Pneumonia **	Pneumonia	8	0.675	0.673	[0.458, 0.876]	−0.002
Pneumothorax **	Pneumothorax	7	0.827	0.721	[0.545, 0.878]	−0.106
Macro average		-	0.820	0.806	-	−0.015

* Label names differ between CheXpert and ChestX-ray14; the closest corresponding class was used. ** Class contains fewer than 10 positive cases.

Across the five pathologies with at least 30 positive cases, the macro-averaged AUC was 0.806 externally against 0.820 for the same classes internally, a difference of 0.015. Performance was thus largely preserved despite a change of institution, imaging equipment, patient population and annotation procedure. Across all seven overlapping classes, the macro-averaged AUC was 0.775; Pneumonia and Pneumothorax contained only 8 and 7 positive cases, and their estimates are not interpretable.

Per-class behavior was heterogeneous, as shown in Figure 7. For Atelectasis and Effusion, the internal estimate falls within the external confidence interval, indicating no detectable change. Consolidation improved from 0.737 to 0.878, plausibly reflecting label quality rather than model behavior: ChestX-ray14 labels were extracted automatically from radiology reports, whereas CheXpert validation labels were assigned by radiologists inspecting the images. Cardiomegaly (0.898 to 0.785) and Edema (0.853 to 0.732) declined; for Cardiomegaly, this may partly reflect the separate Enlarged Cardio mediastinum category in the CheXpert label set. Cardiomegaly has been addressed with dedicated attention-based architectures [54], which may be less sensitive to such label-definition differences, and for edema, the predominantly inpatient case mix. These reductions are discussed further in Section 5. Overall discriminative performance, therefore, transfers to an independent institution, while per-class transferability varies and cannot be inferred from the aggregate value alone.

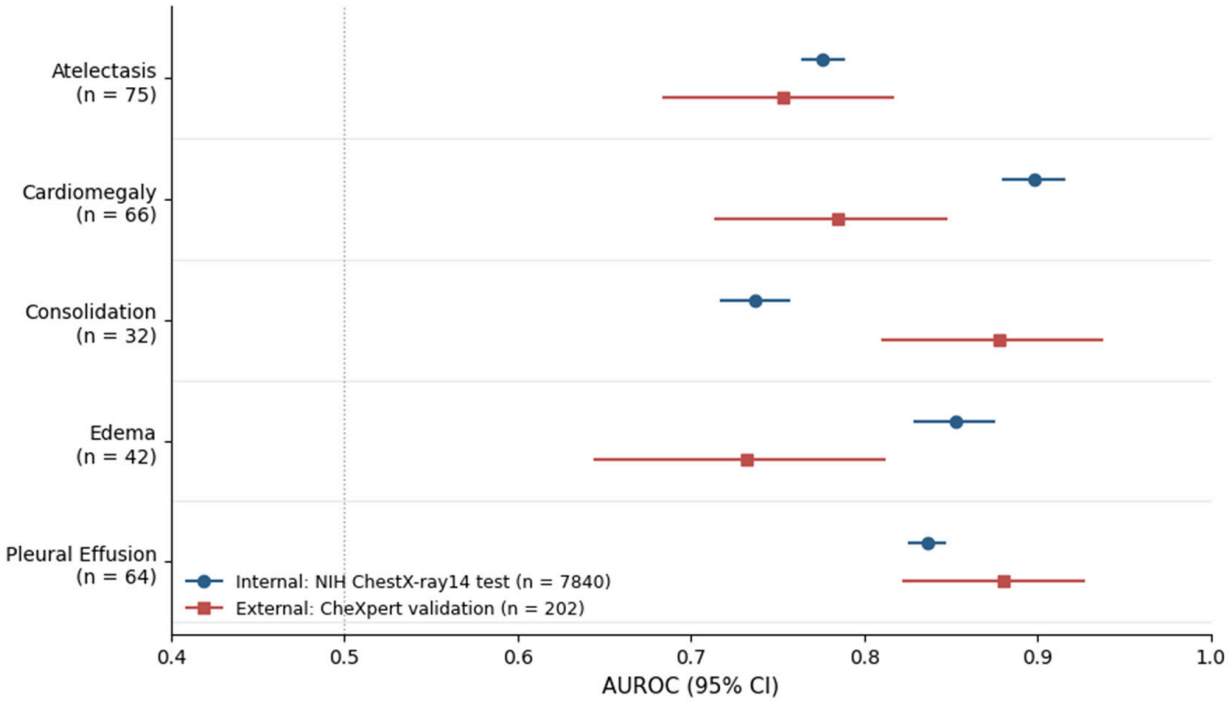


Figure 7. Comparison of internal and external per-class AUROC with 95% confidence intervals for the five overlapping pathologies with at least 30 positive cases.

4.6. Explainability Using Grad-CAM

To improve interpretability of the proposed chest X-ray classification model, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed. Grad-CAM provides visual explanations by highlighting the most important image regions that contribute to a model's prediction. Grad-CAM utilizes the gradients associated with a target class flowing through the network to generate a coarse localization map that highlights the image regions most influential for the model's prediction. The localization map is computed using the gradients of the target concept with respect to the feature maps of the final convolutional layer. This layer is particularly suitable for visualization because it retains rich spatial information while capturing high-level semantic features that are essential for the classification task [55,56]. To demonstrate the interpretability of the proposed model, different cases of positive and negative cases were selected. It should be noted that according to radiographic display conventions, chest X-ray images are presented as if facing the patient; consequently, the image's left and right correspond to the patient's right and left sides, respectively.

Figure 8 shows a true positive pneumothorax case predicted with a predicted probability of 0.994. Grad-CAM activation was concentrated in the right mid-lung field rather than along the pleural margin, where this finding is typically identified.

The Grad-CAM in Figure 9 demonstrates a true positive case of Mass and Grad-CAM activation is centered on a rounded opacity in the right mid-lung field, which corresponds to the region a reader would identify as the lesion. This is one of the clearest spatial correspondences observed among the examined cases.

One of the most clinically meaningful Grad-CAM visualizations is shown in Figure 10. For the cardiomegaly case, activation closely follows the cardiac silhouette, extending to the left cardiac border. Among the cases examined, this showed another clear correspondence between the highlighted region and the anatomical structure relevant to the pathology.

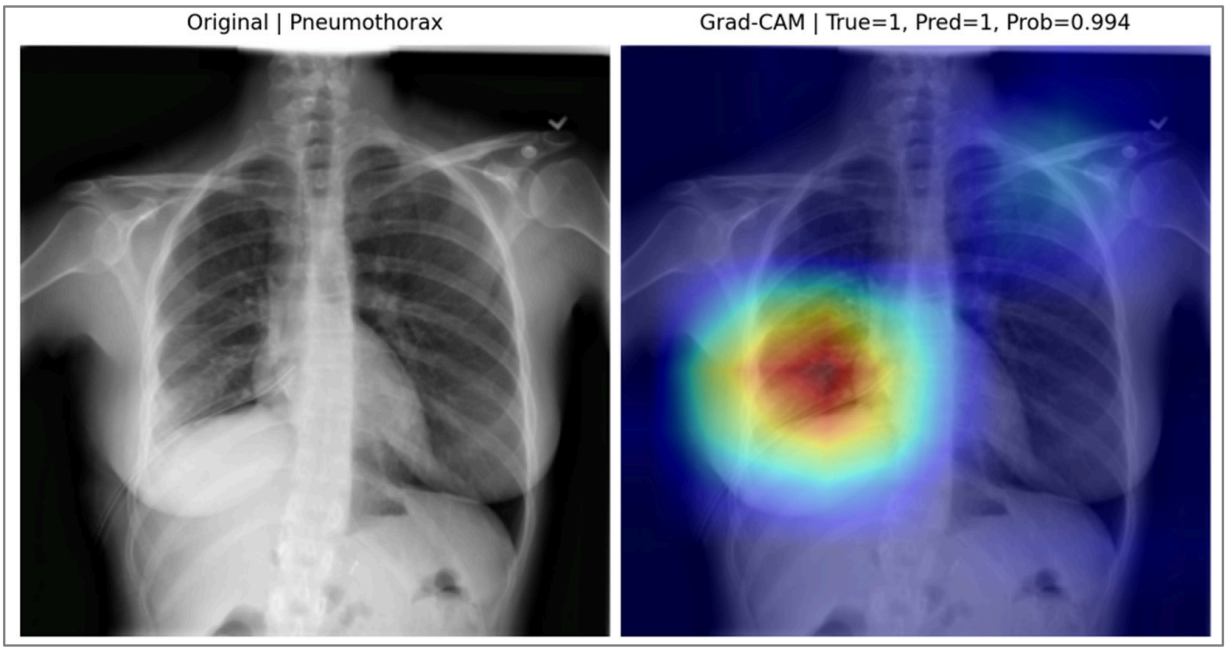


Figure 8. Grad-CAM visualization for Pneumothorax-positive chest X-ray image.

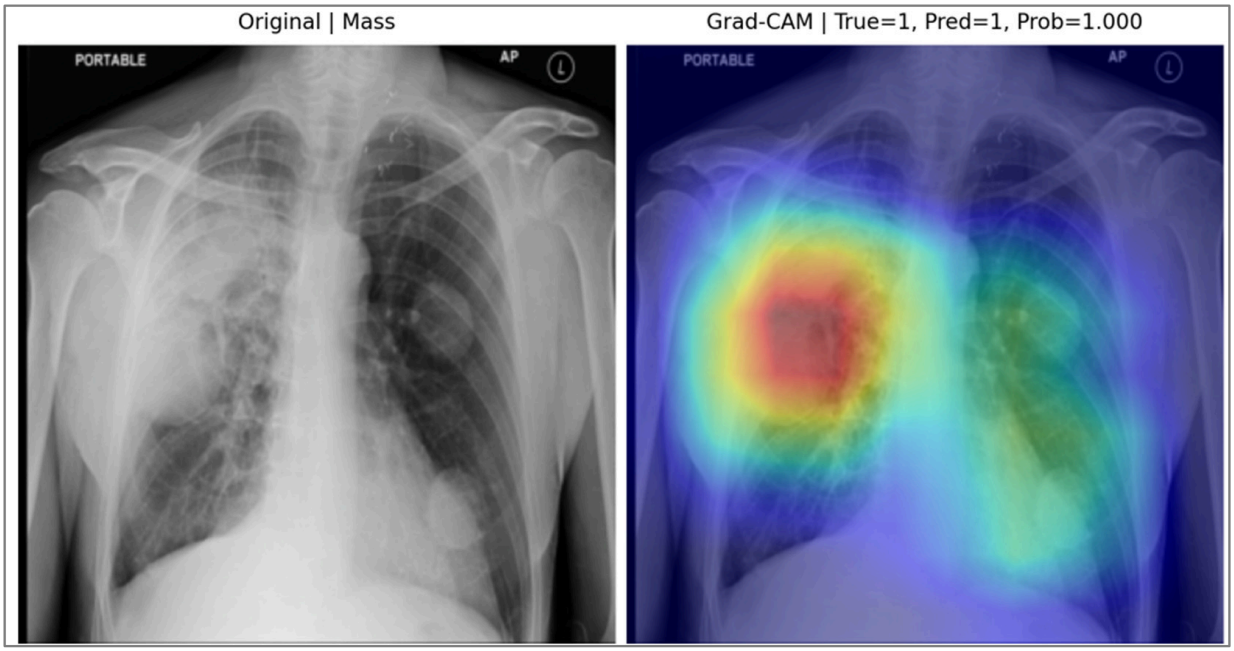


Figure 9. Grad-CAM visualization for Mass-positive chest X-ray image.

The edema case shown in Figure 11 represents a borderline misclassification: the predicted probability of 0.638 fell just below the class-specific threshold of 0.640, and the case was, therefore, assigned a negative label despite a positive reference annotation. Grad-CAM activation was distributed across peripheral regions rather than concentrated in the lower lung fields, where the increased opacity is most pronounced. Such cases illustrate that a small shift in the operating point can change the classification outcome, and that near-threshold predictions should be interpreted with caution.

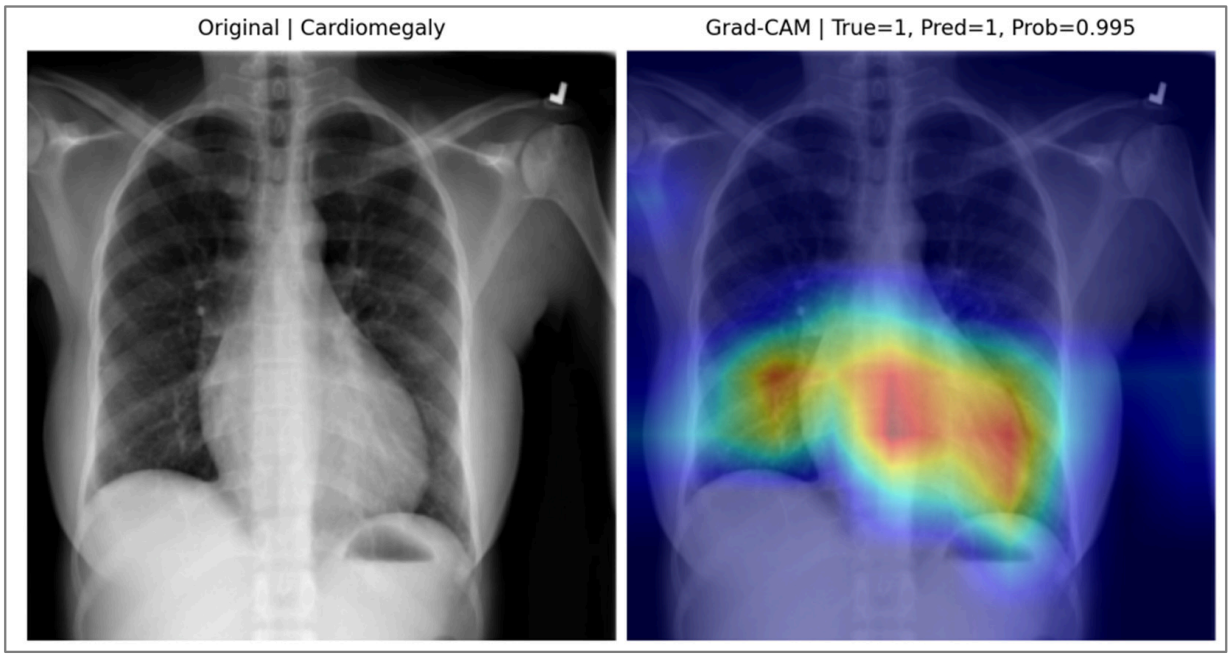


Figure 10. Grad-CAM visualization for Cardiomegaly-positive chest X-ray image.

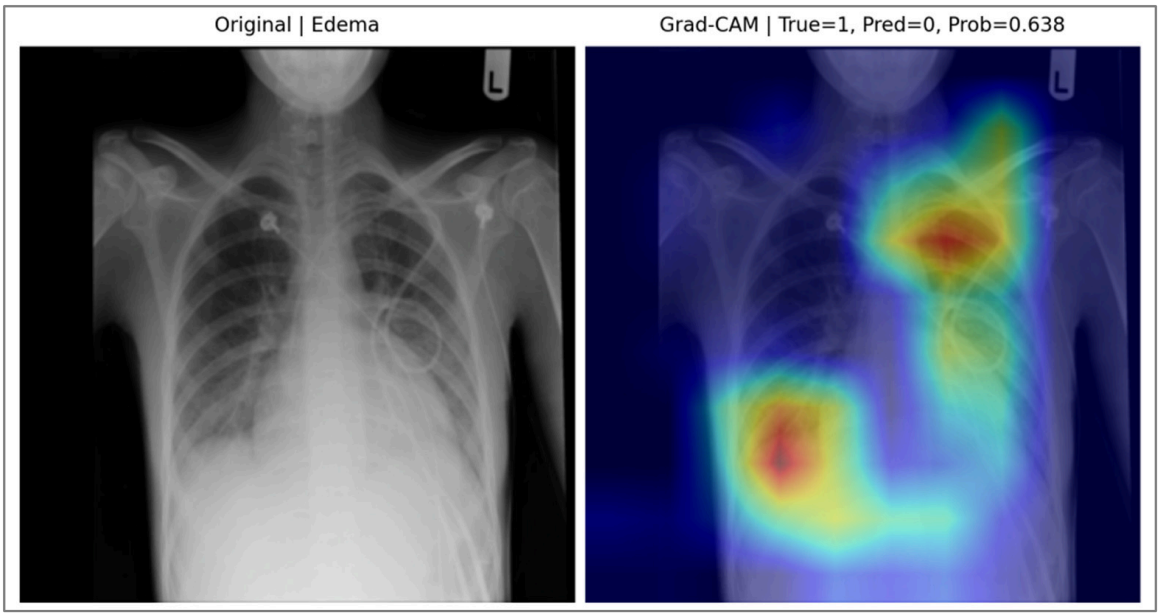


Figure 11. Grad-CAM visualization for a false negative Edema case.

The nodule case shown in Figure 12 was missed by a margin of 0.001 (p = 0.549, threshold 0.550), and the Grad-CAM heat map covers a large portion of the left lung field rather than a focal region, illustrating that class activation maps derived from the final convolutional layer provide limited spatial resolution for small, localized findings such as pulmonary nodules.

The presented Grad-CAM visualizations provide valuable insights into the decision-making process of the proposed DualPool-DenseNet121 model. In several correctly classified cases, the model focused on anatomically relevant lung regions that were consistent with the expected disease locations. The Grad-CAM visualizations provide qualitative insight into the regions associated with individual predictions. For cardiomegaly, activation closely followed the cardiac silhouette, and for the mass case, it was centered on a focal opacity in the right mid-lung field. In contrast, the pneumothorax case was predicted cor-

rectly with high confidence ($p = 0.994$), yet the main activation was located away from the pleural margin, where this finding is typically identified. Spatial correspondence between the highlighted regions and the expected location of the pathology varied considerably across cases and was not related to predictive confidence.

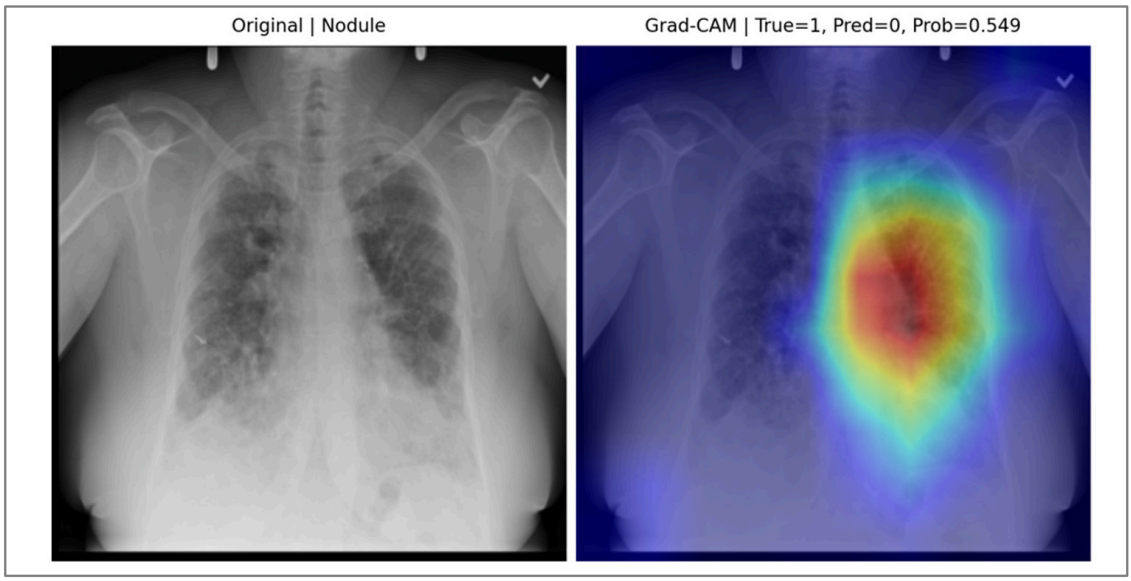


Figure 12. Grad-CAM visualization for a false-negative Nodule case.

The misclassified cases were predominantly borderline predictions below the class-specific threshold, with margins of 0.002 or less for edema and nodule. In these cases, the model assigned probabilities close to the decision boundary, indicating that the outcome was determined by the operating point rather than by a failure to respond to the relevant image content. For the nodule case, activation extended across a large portion of the lung field rather than a focal region, illustrating the limited spatial resolution of class activation maps derived from the final convolutional layer for small findings.

Overall, the Grad-CAM results provide qualitative visual evidence of the regions that contributed most strongly to the model's predictions. It should be emphasized, however, that these interpretability results were not validated by expert radiological assessment; they, therefore, offer preliminary supporting evidence for the plausibility of the model's attention rather than confirmation of clinical correctness. Formal evaluation of the attention maps by radiologists remains an important direction for future work.

5. Discussion

In this study, we evaluated DualPool-DenseNet121 for multi-label classification of thoracic diseases on the NIH ChestX-ray14 dataset. Under strictly patient-wise partitioning, the model achieved a macro-averaged AUC of 0.803 and a micro-averaged AUC of 0.831, with the strongest per-class results for Emphysema (0.916), Cardiomegaly (0.898) and Hernia (0.885). The micro-averaged value exceeds the macro-averaged one because it is dominated by the most prevalent pathologies, on which the model performs better, whereas the macro average weights rare classes such as Hernia equally with common ones.

Three findings merit particular attention. First, the ablation study showed that combining average and maximum pooling does not confer a statistically robust advantage over global average pooling alone, with macro-averaged AUC differing by at most 0.004 between the three variants. The improvement over the baseline is, therefore, attributable to the training pipeline rather than to the pooling head. Second, the same model and protocol yielded 0.858 under image-level partitioning and 0.803 under patient-wise partitioning.

This difference of 0.055 exceeds the difference between any of the published results compared in Table 6 and indicates that reported performance differences of this magnitude cannot be interpreted without knowing how the data were partitioned. The same concern has been raised independently by Hanif et al. [36], who note that studies not using the official split of this dataset report higher average AUC and that such comparisons must be interpreted with caution. Third, zero-shot external validation on the CheXpert validation set yielded a macro-averaged AUC of 0.806 against 0.820 for the same five classes internally, indicating that discriminative performance is largely preserved across institutions, acquisition equipment and annotation procedures.

The Grad-CAM analysis provided qualitative insight into the regions associated with individual predictions. Spatial correspondence between activation and the expected location of the pathology varied considerably and was not related to predictive confidence: for cardiomegaly and mass, activation coincided with the relevant structures, whereas the pneumothorax case was predicted correctly with high confidence yet showed activation away from the region where this finding is typically identified. The misclassified cases examined were predominantly borderline predictions falling marginally below the class-specific threshold, indicating that the outcome was determined by the operating point rather than by a failure to respond to the relevant image content. These observations were not verified by a radiologist and are presented as preliminary interpretability evidence.

Several limitations remain. The model performed less well on Pneumonia, Infiltration, Consolidation and Pleural Thickening, consistent with known characteristics of ChestX-ray14, including class imbalance, noisy labels derived by automated extraction from radiology reports, and overlapping radiographic presentations. The external evaluation was based on a single cohort of 202 radiographs from one institution, whereas multicenter evaluations of chest radiograph models have been reported on substantially larger cohorts [57], and two classes, Cardiomegaly and Edema, showed reduced performance in that setting. The Grad-CAM visualizations were not assessed by expert radiologists, and no calibration analysis or prospective evaluation was performed. Future work will address class-balanced loss functions, targeted augmentation for under-represented pathologies, calibration analysis, and evaluation on additional external cohorts such as MIMIC-CXR and VinDr-CXR. Transformer-based architectures, which have recently been applied to chest radiograph classification [58], represent a further direction for comparison under the same evaluation protocol.

6. Conclusions

This study evaluated DualPool-DenseNet121, a DenseNet-121 architecture with a dual-pooling classification head, for multi-label classification of fourteen thoracic pathologies on the NIH ChestX-ray14 dataset. Under strictly patient-wise partitioning with verified zero patient overlap, the model achieved a macro-averaged AUC of 0.803 and a micro-averaged AUC of 0.831, together with per-class threshold-based metrics obtained at operating points selected on the validation set.

An ablation study showed that combining average and maximum pooling does not provide a statistically robust advantage over global average pooling alone. The improvement over the baseline is, therefore, attributable to the pipeline, comprising staged domain-specific initialization, targeted augmentation of under-represented classes and per-class threshold selection, rather than to the pooling head in isolation.

Evaluating the same model under two partitioning strategies yielded a difference of 0.055 in macro-averaged AUC. This exceeds the differences reported between many published methods on this dataset and indicates that comparisons across studies are informative only when the partitioning strategy is matched and explicitly stated.

Zero-shot external validation on the CheXpert validation set indicated that discriminative performance is largely preserved across institutions, with a macro-averaged AUC of 0.806 against 0.820 internally for the same five pathologies, although per-class transferability varied.

Taken together, these results provide a reproducible retrospective benchmark for multi-label thoracic abnormality classification. Expert radiological assessment, calibration analysis and prospective evaluation are required before any clinical application can be considered.

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